

MAT 115E Introduction to Programming Language

Lab-9/ CRN : 23492

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1 Question 1

Write a C program that accomplishes the followings:

- Create an array of size N **dynamically**. The size (N) of array must be read from user.
- This array should be filled with the following formula.

$$a_i = \begin{cases} (19 * i + 73) \pmod{N}, & \text{if array index } i \text{ is even} \\ (13 * i + 94) \pmod{N}, & \text{if array index } i \text{ is odd} \end{cases}$$

- Compute average of numbers in the array and assign this value to the variable named **averageFirstArray**.
- The size of array you created should be dynamically expanded to size of $2 * N$
- Analogously, you filled the expanded parts of array with the following formula.

$$a_i = \begin{cases} (7 * i + 119) \pmod{2N}, & \text{if array index } i \text{ is even} \\ (11 * i + 29) \pmod{2N}, & \text{if array index } i \text{ is odd} \end{cases}$$

- Calculate average of all numbers in the array of size $2 * N$ and assign this value to the variable named **averageExpandedArray**.
- Determine in which case the array has the maximum average. You can write simple message as *"Array has maximum average before expanded"* or *"Array has maximum average after expanded"*.
- Lastly, spaces allocated in memory must be freed.

2 Question 2

Write a C program that implements a simple student grade management system using dynamic memory allocation.

Your program must perform the following steps:

- Ask the user how many students' grades will be entered initially.
- Allocate a **dynamic array** to store student grades.
- Take the grades from the user and store them in the array.
- Then ask the user how many additional student grades will be added.
- Resize the array according to the new number of students.
- Take the additional grades from the user and store them in the expanded part of the array.
- Finally:
 - Print all grades.
 - Calculate the average of the grades.
 - Find the number of grades that are greater than the class average.
 - Find the highest and lowest grades.
- At the end of the program, free all dynamically allocated memory.